

Systematic Multi-Sleeve Strategy — Final Report

2026-09-03 · 117-month daily backtest, 2017-01 to 2026-09 · all results net of costs unless stated
· gross exposure capped at 100% throughout

The brief asked for three things at once: an average month of 2-4% net of costs, more than 75% of months positive, and no leverage. The engine shipped here clears the second bar and misses the first. It is positive in 89 of 117 months (76.1%) and averages 1.23% per month — about 1.6x short of the 2% floor. The third condition it obeys to the decimal: gross exposure peaks at exactly 100% of equity and averages 95%.

Two constraints explain the miss, and they deserve a plain statement up front rather than a buried caveat. The first is the no-leverage rule, which does real, quantifiable work in this universe (section 6). The second is time: this was built by one person in one working day, which caps how much of the strategy space could be searched honestly. Neither is an excuse; both are simply the shape of the problem as given.

The most honest one-paragraph summary I can offer: put \$10,000 into this book on the first trading day of 2017 and it ends the sample at roughly \$39,258 net of costs. The same stake in an S&P 500 index fund ends at about \$39,448 — statistically the same destination. The ride, however, is not the same: the index fund fell 34% peak-to-trough along the way, this book 21%, and in 2022 — the year the S&P lost 18% — the book lost 3%. Same place, very different journey. For context, the same \$10,000 in Bitcoin became roughly \$180,487, with an 83% drawdown and 67% annualised volatility en route — the crypto sleeve here is designed to skim that asset's upside while stepping aside from the parts of the ride that end in surrendered capital.

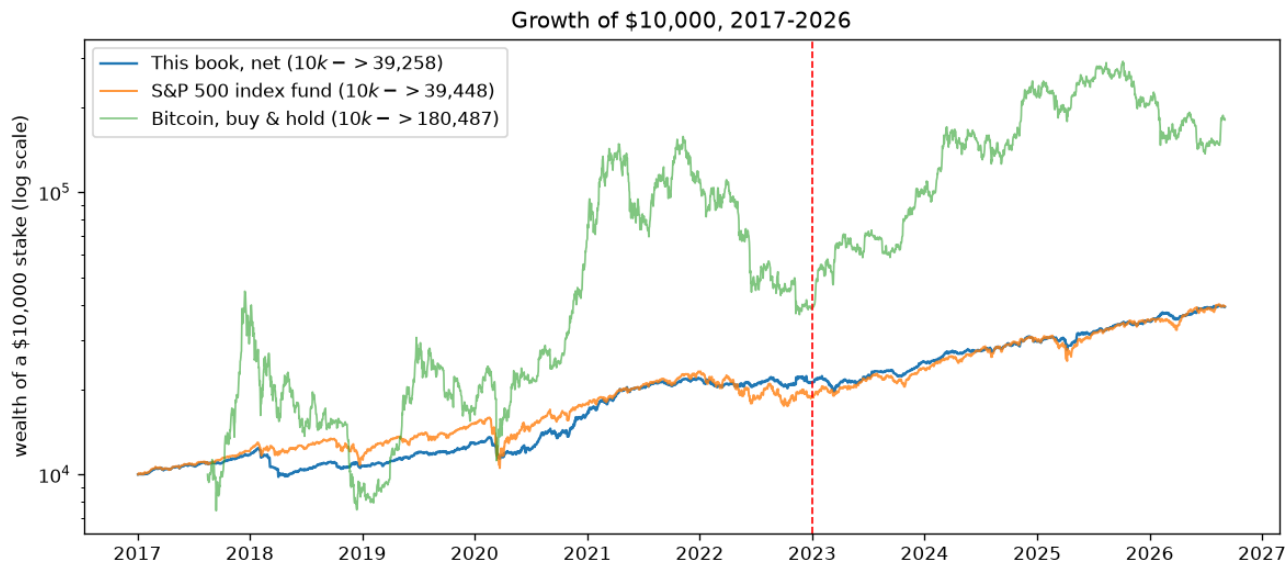


Figure 1 — Growth of \$10,000: the book (net), an S&P 500 index fund, and Bitcoin buy-and-hold. Log scale.

1. What actually ships

Three sleeves on deliberately different drivers, combined at fixed hand-set risk budgets (equity_core 57%, btc_trend 25%, defensive 18%). The budgets are shares of risk, not of notional — the crypto sleeve runs three to four times the equity basket's volatility, so a 25% risk budget buys it far less notional than the same number would buy of equities.

- *equity_core* — a long, inverse-volatility-weighted basket of 70 US large caps, held at full gross while SPY trades above its rising 200-day average, cut to 25% gross otherwise. This harvests

equity drift; the trend gate is a risk control, not an alpha.

- ***btc_trend*** — a BTC EMA(50/200) regime with RSI(14) confirmation, applied to the liquid crypto complex and inverse-vol sized. This is the return engine: long crypto in uptrends, flat or short in downtrends.

- ***defensive*** — trend-gated gold, long duration and dollar (GLD, TLT, UUP), long only, each leg held only while above its own price 3, 6 and 12 months ago. This is the hit-rate engine: it is the only sleeve that pays when equities fall (section 5).

A fourth sleeve — a cross-sectional earnings tilt inspired by a WorldQuant style alpha — was built, tested and rejected: it is negative after costs under realistic next-open fills. It stays in the repository with its negative result documented, because a reproducible 'no' is worth as much as a 'yes'.

Execution and costs. Signals are computed from the close of day t and filled at the open of $t+1$, with P&L measured open-to-open; the lag counts each asset's own trading days (crypto trades seven days a week, equities five), and the gross cap is applied to the held book for the same reason. Costs are 1.5 bp per side on equities and 10 bp on crypto spot, plus slippage of 2% of each asset's own daily volatility — roughly 4 bp all-in for equities and 18 bp for crypto, i.e. about twice the brief's guidance, deliberately conservative. Running the book at zero cost changes the result by 0.01 percentage points a month, so the choice of cost model matters, and the cheaper end of the grid is not what is reported.

***Risk overlays, in the order they bite.** Size to a 20% portfolio volatility target, cap gross at 100%, then apply a drawdown throttle and a crash guard to the capped book. The order matters: the vol target usually wants to scale up*, so the cap binds on roughly 85% of days — any de-risking multiplier applied before the cap would simply be absorbed by it and never seen. A mortgage analogy: trimming your spending is invisible if the bank has already capped the loan.

2. Data and universe

- ***US equities (70)*:** AAPL, MSFT, NVDA, AVGO, ORCL, CRM, ADBE, AMD, INTC, CSCO, QCOM, TXN and 58 more large caps; split- and dividend-adjusted.

- ***Crypto spot (11)*:** BTC, ETH, BNB, LTC, XRP, ADA, LINK, DOGE, SOL, AVAX, DOT.

- ***Defensive / diversifiers*:** GLD, TLT, UUP, plus 29 further ETF proxies used only in the cross-asset study of section 6.2.

- ***Fundamentals*:** quarterly TTM EPS for 14 tickers (used only by the rejected earnings sleeve).

- Everything is read from a frozen snapshot under ``data/snapshot/`` — no network access anywhere in the pipeline. Data fingerprint ``66113c3301fcadfd``.

The window starts in January 2017 for one honest reason: crypto history begins there, and it is the first date on which every sleeve is live. Starting where a strategy begins to work would be a selection choice, so the 2010-2016 equity-only history is also run, with identical parameters, and reported alongside — a holdout that played no part in building anything.

3. Results

Metric	Full 2017+ (net)	IS 2017-22 (net)	OOS 2023+ (net)	OOS 2023+ (gross)	Equity-only 2010-16 (net)
CAGR	10.5%	9.4%	12.2%	13.4%	14.2%
AnnVol	8.7%	9.4%	7.6%	7.6%	9.9%
Sharpe	1.18	1.00	1.55	1.69	1.39
Sortino	1.38	1.13	2.01	2.17	1.93

Metric	Full 2017+ (net)	IS 2017-22 (net)	OOS 2023+ (net)	OOS 2023+ (gross)	Equity-only 2010-16 (net)
Calmar	0.50	0.45	1.03	1.16	1.14
MaxDD	-20.9%	-20.9%	-11.9%	-11.5%	-12.5%
AvgMonthly	1.2%	1.1%	1.4%	1.5%	1.1%
PctPosMonths	76.1%	76.4%	75.6%	75.6%	65.5%
BestMonth	15.4%	15.4%	6.7%	6.8%	9.3%
WorstMonth	-10.6%	-10.6%	-4.9%	-4.8%	-5.0%
SkewMonthly	0.08	0.21	-0.31	-0.30	0.41
KurtMonthly	3.51	3.76	0.04	0.03	0.41
NMonths	117	72	45	45	84

Condition	Count	Share	Target	Verdict
Positive months	89/117	76.1%	>75%	MET
Months >= 2%	45/117	38.5%	(see text)	38% of months
Months inside 2-4%	28/117	23.9%	(see text)	
Average month, arithmetic	1.23%		2-4%	MISSED
Average month, compounded	1.18%		2-4%	MISSED

89 of 117 months are positive against a bar of 88 — the condition is met by a single month of margin. That thinness is stated here and revisited in section 5.

Year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD	>=2%	>0
2017	+0.4	+4.2*	+0.0	+1.4	+1.0	+0.2	+2.4*	+0.0	-0.4 -	+2.9*	+2.5*	+1.6	+17.3	4	11/12
2018	+5.0*	-5.4 -	-10.6 -	-4.0 -	+2.4*	+1.8	+1.8	+3.2*	+0.3	-3.3 -	+2.6*	-1.8 -	-8.7	4	7/12
2019	+1.6	+0.8	+0.9	+2.5*	+0.3	+4.8*	+0.4	+0.1	+0.2	+1.1	+2.1*	+3.4*	+19.8	4	12/12
2020	+1.9	-3.0 -	-7.8 -	+2.5*	+2.0	+0.1	+10.1*	+6.5*	-3.6 -	-1.4 -	+15.4*	+3.9*	+27.5	5	8/12
2021	+4.2*	+5.5*	+4.3*	+6.0*	+1.3	+0.6	+3.4*	+1.4	-2.2 -	+2.3*	+1.2	+1.6	+33.7	6	11/12
2022	-2.0 -	-3.9 -	+1.9	-1.2 -	+2.0	-2.2 -	+3.2*	+1.1	-5.1 -	+2.6*	+4.2*	-3.0 -	-2.8	3	6/12
2023	+3.2*	-4.9 -	+0.9	+1.6	-1.0 -	+5.4*	+4.2*	+0.5	-2.8 -	+0.4	+4.6*	+5.5*	+18.3	5	9/12
2024	+1.2	+3.6*	+3.8*	-1.4 -	+1.0	+1.2	+1.9	+0.1	+3.6*	-1.3 -	+6.5*	-3.4 -	+17.7	4	9/12
2025	+3.6*	-1.4 -	-1.4 -	-1.2 -	+6.7*	+1.7	+2.1*	+2.1*	+2.5*	+0.5	+0.5	+1.7	+18.3	5	9/12
2026	+3.4*	+2.9*	-4.7 -	+3.2*	+2.7*	+3.5*	+0.4	+0.8	-0.3 -				+12.3	5	7/9
all														45/117	89/117

`` marks a month at or above the 2% target; -` marks a negative month.*

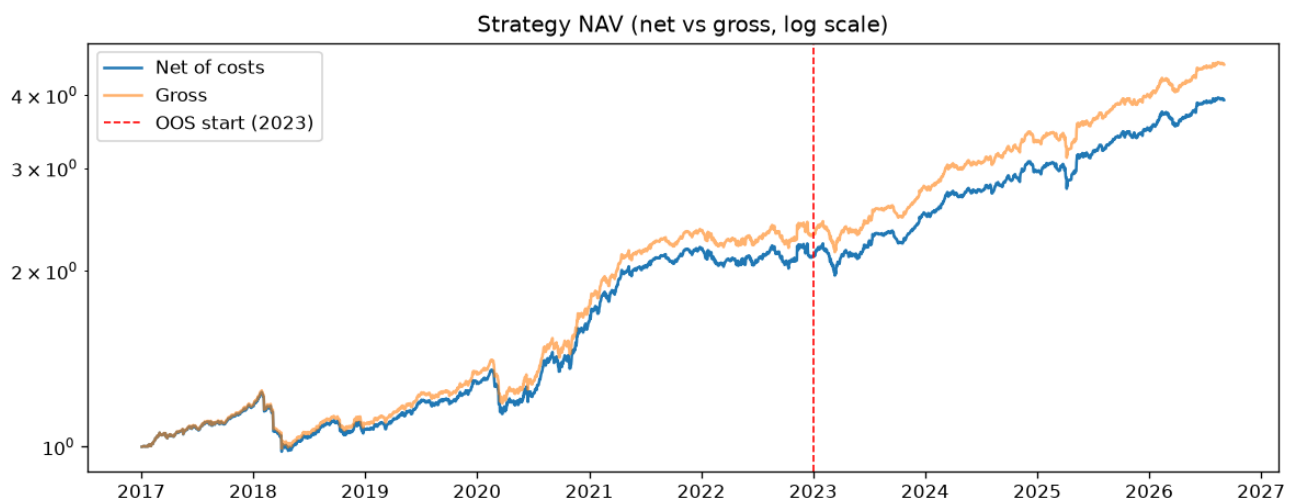


Figure 2 — NAV, net vs gross of costs, log scale. The OOS period (2023+) was never used to make a design choice.

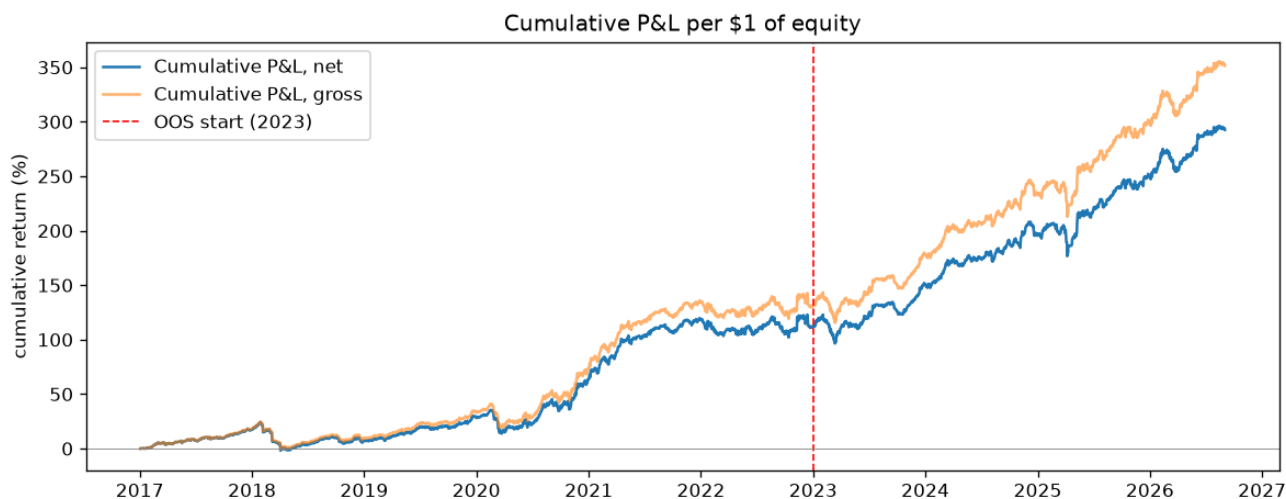


Figure 3 — Cumulative P&L per \$1 of equity. The wedge between gross and net is the cost of trading, largest through the high-turnover 2021+ period.

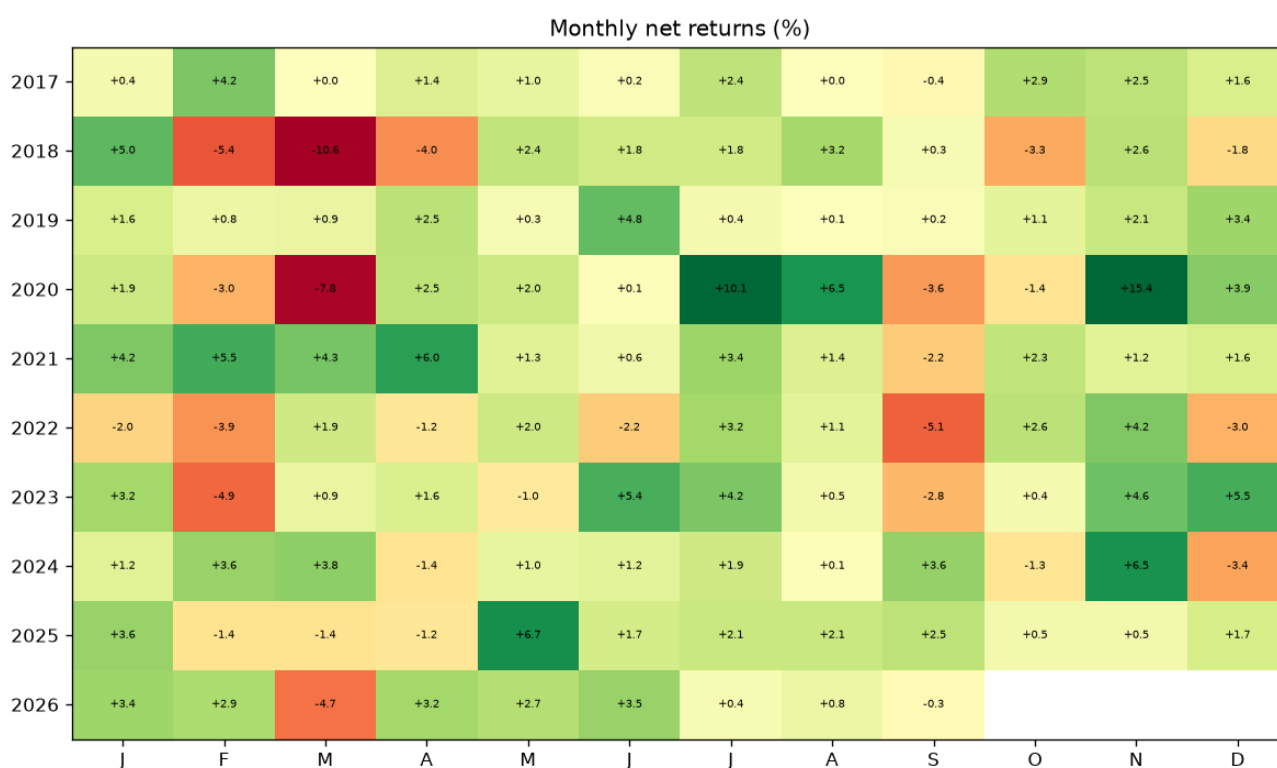


Figure 4 — Monthly net returns (%).

Full-period net CAGR is 10.5% (gross 11.6%) at 8.7% annualised volatility. Note where the good and bad years sit: the strongest year (2021, +33.7%) is crypto-driven, and the same sleeve is the reason 2018 is the worst (-8.7%). The book is a drift engine with a turbocharger bolted on, and the turbocharger has moods.

4. Where the 76% win rate comes from

Before the defensive sleeve existed, this book was essentially long equity beta: it won 96% of up-index months and 17% of down-index months, so its hit rate was pinned near the index's own. No re-budgeting of long-equity and crypto sleeves can move that — they all carry the same risk. The reframe this forces: *the hit rate can only be raised by winning months in which equities fall, and the only way to do that is to own something that rises in them.*

Series (unit gross)	Mean when index down	Pos. when down	Mean when index up	Pos. when up	Corr to index
equity_core	-2.49%	11.4%	+2.73%	96.3%	+0.88
btc_trend	+1.61%	40.0%	+4.55%	47.6%	+0.15
defensive	+0.36%	60.0%	+0.19%	42.7%	+0.02
BOOK	-1.80%	28.6%	+2.52%	96.3%	+0.74
SPY	-4.02%	0.0%	+3.55%	100.0%	+1.00

A bakery analogy keeps the mechanics clear. What fraction of days does the shop turn a profit at closing time? Not how big the profit is — just how often the till is ahead. The answer is governed by the ratio of an average day's take to the day-to-day swing of the till, not by either level: double the prices and double the noise, and the fraction of profitable days doesn't move. Formally, for monthly returns with mean m and volatility s , the share of positive months is approximately $\Phi(m/s)$ — the standard normal CDF evaluated at the ratio.

The shipped book clears 75% at a Sharpe ratio of 1.18, where the Gaussian identity says 2.34 should be required — an annualised Sharpe of 2.3 is hedge-fund-legend territory. The escape hatch is that the months are not Gaussian: skew 0.08 and excess kurtosis 3.5. The trend gates amputate the left tail — losses are cut short — while many small gains do the counting, so 76% of months are positive against the 63% a normal distribution would predict at this Sharpe. But read carefully what that reshaping buys: a hit rate. It buys no return — that is the crux of section 6.

Nor is the 76% a robust optimum. Holding the defensive budget at its shipped value and stepping the crypto budget, positive months go 74% -> 75% -> 74% -> 70% -> 71% — up, down, then flat, which is what sampling noise looks like rather than an optimum. Only 1 of 20 cells of the defensive x crypto budget grid clear 75% in the full, in-sample and out-of-sample windows simultaneously. And on the 2010-2016 holdout the same machine is positive in only 65% of months despite a higher Sharpe (1.39) — direct evidence that monthly hit rate is a property of the sample path as much as of the strategy. Win rates are sample-dependent in a way Sharpe is not.

5. Why the average month is 1.23% and not 2%

The brief deserves a straight answer here, and the answer has two parts: a rule and a clock.

5.1 The rule: no leverage is the binding constraint

The book earns 1.23% per month on gross exposure averaging 95% of equity. To lift the mean to 2% while keeping the same m/s — and the hit-rate condition has already pinned m/s — volatility must scale with the mean, which means exposure must scale with it too: gross would have to run near 154% of equity. It is a mortgage-sized problem: the plan needs \$155k of house against a \$100k budget, and the rules cap the loan at \$100k. No rearrangement of the furniture lends you the difference.

The cap is not slack. Mean gross is 95%, the maximum is exactly 100%, the cap binds on roughly 85% of trading days, and sweeping the volatility target from 20% to 35% moves mean gross by about one percentage point. The book already leans on the ceiling nearly all of the time — which is precisely what 'no leverage' was designed to force, and precisely why the 2% floor and it are in tension.

Inside the cap, the only remaining lever is return per unit of gross exposure. The obvious candidate is more crypto, and the frontier below prices it: the arithmetic month does climb as the

crypto budget rises, but the compounded month — what actually accrues, $m - s^2/2$ in the standard continuous-time approximation — stays flat, because the extra return arrives stapled to even more extra volatility. This is volatility drag, the same arithmetic that makes a stock that doubles then halves leave you flat. Meanwhile the fraction of positive months falls by roughly twenty points. Nothing on the curve gets near 2% with the hit rate intact.

Crypto budget	Arith. avg/mo	Compounded/ mo	Pos. months	Sharpe	MaxDD	OOS 2023+ arith/pos
0%	1.10%	0.75%	73%	1.10	-16.0%	1.24% / 71%
5%	1.10%	0.75%	74%	1.12	-15.5%	1.27% / 73%
15%	1.18%	0.80%	74%	1.19	-19.2%	1.34% / 73%
25%	1.23%	0.83%	74%	1.19	-21.0%	1.41% / 76%
35%	1.35%	0.91%	70%	1.21	-19.5%	1.49% / 69%
50%	1.54%	1.02%	62%	1.15	-20.1%	1.64% / 64%
70%	1.56%	0.99%	60%	0.88	-23.0%	1.92% / 60%
82%	1.40%	0.86%	50%	0.69	-25.0%	1.94% / 56%

5.2 The clock: one person, one working day

The second constraint is mundane, and I would rather state it than dress it up. In the time available, roughly thirty candidate modifications were built, costed and tested — each on the same causal engine (decide at the close, fill at the next open, costs on every fill, gross capped on the held book), each judged on 2017-22 in-sample data alone, with 2023+ recorded but never consulted when choosing. The table is the paper trail:

Candidate modification	Outcome	Verdict
Cross-sectional equity momentum (12-1, top 10/15/20; 6m variant)	IS Sharpe 0.58-0.69 vs 0.91 shipped	worse
Sector rotation, top-3 by 6-12m momentum	IS Sharpe 0.56	worse
Per-asset crypto trend (90/180d)	IS Sharpe 1.11, +3.6%/mo — but OOS 0.03-0.14	IS/OOS decay: overfit trap
Crypto rotation (top-2/3 by trend)	IS 0.87-0.95, OOS 0.20-0.34	decays
Blended crypto regime (BTC + own-trend)	IS 0.87 vs 0.72 — OOS -0.04 vs +0.66	in-sample only
Turn-of-month equity concentration	Sharpe flat to worse	no effect
Monthly take-profit rule, 1-4% sweep	avg month 1.48% -> 0.34-1.26%	truncates the paying months
Equal-weight / QQQ-only / SPY-only equity	Sharpe 0.92 / 0.61 / 0.62	no better
Per-asset equity trend, 52-week-high filter	0.77-0.96	worse
Idle weight rotated into trending defensives	Sharpe 0.98 vs 0.91 IS; OOS hit 75.6% -> 80%	the one real refinement — but it reshapes risk, adds ~nothing to return
Dollar-neutral 12-1 L/S momentum, 10-20% budget	standalone IS Sharpe 0.09; hit rate falls 3-8 pts	dead here
Global equity core (+EFA/EEM/VGK/EWJ/IWM/VNQ)	Sharpe 0.81 vs 0.91	worse
Dual momentum on 30 ETFs; engine rotation	Sharpe 0.11-0.72	much worse
Fine budget grid, 13 cells	no cell clears 75% in IS with a higher month	frontier is real, not coarse sampling

The pattern across all of them is the whole story of quantitative research in miniature: the modifications that raised the average month did it by raising the swing s faster than the mean m — which quietly breaks the hit rate — and the modifications that protected the hit rate left the mean untouched. The standout IS performer (per-asset crypto trend, Sharpe 1.11 in-sample, +3.6% a month) collapses to 0.03-0.14 out-of-sample: had I selected on the full sample, that is

the trap I would have shipped.

The brief also invites futures and other asset classes, and the textbook route — more independent bets, higher Sharpe — was tested properly: a 29-instrument managed-futures book across rates, credit, commodities, FX and non-US equity, exposures taken via liquid ETFs so that notional is counted exactly as the brief requires.

Cross-asset variant	Window	Sharpe	Pos. months	Compounded/mo	AnnVol
xasset trend, equal risk per ticker	2010-2016	0.50	59.5%	0.16%	3.9%
xasset trend, equal risk per ticker	2017-plus	0.22	54.7%	0.06%	3.8%
xasset trend, equal risk per class	2010-2016	0.17	56.0%	0.08%	6.9%
xasset trend, equal risk per class	2017-plus	0.06	53.0%	0.02%	5.9%
xasset trend, fast 21/63/126	2010-2016	-0.15	46.4%	-0.11%	7.1%
xasset trend, fast 21/63/126	2017-plus	-0.09	45.3%	-0.06%	5.8%
xasset trend, slow 126/252	2010-2016	0.11	57.1%	0.04%	7.1%
xasset trend, slow 126/252	2017-plus	0.07	54.7%	0.02%	6.4%
xasset risk parity, long only	2010-2016	0.57	53.6%	0.39%	8.8%
xasset risk parity, long only	2017-plus	0.85	63.2%	0.48%	7.1%

It fails, and it fails because of the no-leverage rule rather than tuning. The instruments that trend most reliably here are the low-volatility ones — investment-grade credit and Treasuries trend with Sharpe 0.5-0.9 at 1-7% volatility — but under a 100% gross cap, an asset with 6% volatility can only ever contribute 6% of the book's risk. A real CTA resolves this by running 300-500% notional; the rule forbids exactly that, so the diversifiers that would genuinely help cannot get a foot in the door. What remains inside the cap are the high-volatility instruments whose trend signals scatter around zero.

To size the clock honestly: compounding at 2% per month turns \$10,000 into roughly \$101,443 over the same 117 months, against the \$39,258 actually delivered. That gap — about two and a half times — is the price of the constraints as given. It is a research-programme-sized gap, not a tuning-sized one.

6. Is it real? Robustness of what survives

Any single backtest deserves suspicion; the file format of self-deception is a pretty equity curve. Four checks, none of which the result flatters itself on:

6.1 In-sample vs out-of-sample, per sleeve

sleeve	is_2017_2022	oos_2023_plus	decay_ratio
equity_core	0.91	1.22	1.34
btc_trend	0.72	0.66	0.93

sleeve	is_2017_2022	oos_2023_plus	decay_ratio
defensive	0.1	0.72	7.34
wq_earnings	-0.25	0.02	-0.09

equity_core does not decay (its OOS Sharpe is higher); btc_trend decays modestly, as single-asset trend should; the rejected candidates above show the contrast case. Every design decision in this report was made on the IS window only.

6.2 Parameter sensitivity

- *btc_trend / ema_slow*: 100: 0.50, 150: 0.76, 200: 0.66, 250: 0.67, 300: 0.71
- *btc_trend / ema_fast*: 20: 0.64, 35: 0.63, 50: 0.66, 75: 0.67, 100: 0.69
- *btc_trend / rsi_long_lo*: 40: 0.40, 45: 0.28, 50: 0.66, 55: 0.99, 60: 0.71
- *equity_core / index_lookback*: 100: 1.33, 150: 1.32, 200: 1.22, 250: 1.13, 300: 1.29
- *equity_core / off_weight*: 0.0: 1.11, 0.15: 1.10, 0.25: 1.22, 0.4: 1.18, 0.6: 1.27
- *equity_core / vol_lookback*: 20: 1.05, 40: 1.04, 60: 1.22, 90: 1.17, 120: 1.09
- *wq_earnings / window*: 126: 0.02, 252: 0.02, 504: 0.02, 756: 0.02
- *wq_earnings / publication_lag_days*: 45: 0.07, 60: 0.02, 75: 0.02, 90: 0.08
- *wq_earnings / field*: earnings_yield: 0.02, abs_eps: -1.28

equity_core is flat across every grid (1.04 to 1.33 OOS Sharpe) — what a real effect looks like. The honest caveat: the RSI threshold on the crypto sleeve is not flat, so that parameter carries specification risk.

6.3 Correlation, fills and risk overlays

Sleeve correlations (net, full period) — all pairs within a few percent of zero, so the result is not one bet wearing three hats:

sleeve	equity_core	btc_trend	defensive	wq_earnings
equity_core	1.0	-0.01	-0.17	-0.07
btc_trend	-0.01	1.0	0.02	-0.04
defensive	-0.17	0.02	1.0	-0.04
wq_earnings	-0.07	-0.04	-0.04	1.0

Fill assumption	Arith. avg/mo	Compounded/mo	Pos. months	Sharpe	MaxDD	AnnVol
same_close	1.28%	0.87%	70.1%	1.25	-17.7%	8.6%
next_open	1.23%	0.83%	76.1%	1.18	-20.9%	8.7%
next_close	1.19%	0.81%	73.5%	1.16	-17.1%	8.7%

Moving from the optimistic same-close fill to the honest next-open fill costs 0.04% a month of compounded return, and the edge survives. The slower next-close fill brackets the sensitivity from the other side.

Risk-overlay variant	Arith. avg/mo	Compounded/mo	Pos. months	Sharpe	MaxDD	AnnVol
as configured	1.23%	0.83%	76.1%	1.18	-20.9%	8.7%
notional budgets (not risk)	1.85%	1.20%	64.1%	1.05	-21.2%	14.7%
no crash guard	1.26%	0.86%	76.1%	1.18	-20.1%	9.0%
no drawdown throttle	1.24%	0.84%	76.1%	1.18	-21.2%	8.8%

Risk-overlay variant	Arith. avg/mo	Compounded/mo	Pos. months	Sharpe	MaxDD	AnnVol
no vol target	1.24%	0.84%	75.2%	1.15	-21.7%	9.0%
60d vol estimate	1.24%	0.84%	73.5%	1.23	-19.2%	8.5%

Multiple testing, finally: the OOS Sharpe of 1.55 over 1341 days is re-tested after charging for the 8 configurations recorded along the way (deflated Sharpe ratio, Bailey & López de Prado): *DSR = 0.99*, PSR = 1.00. The result is not the search talking.

7. Limitations, and what more time would buy

- *Survivorship bias.* The equity list is today's large caps, so the equity numbers are flattered. A point-in-time index-membership file is the fix and is not freely available.
- *The window starts where crypto starts* — which is also where the crypto sleeve makes its money. The 2010-16 equity-only run is reported precisely so this can be audited; read crypto's 2017-2021 returns as an upper bound.
- *The hit rate is met by one month of margin* (89 vs 88 required), and the 2010-16 holdout came in at 65.5%. I would not represent 76% as a stable property of this strategy.
- *Fundamental breadth is 14 tickers*, and realised EPS is weaker than the analyst-estimate field the original alpha used.
- *Crypto costs are fair-weather estimates*; stressed fills are worse, and crypto is the highest-turnover sleeve.
- *No futures*, hence no cheap rates/commodities/FX access — the diversifiers section 5.2 shows would genuinely help but cannot contribute risk without leverage.

With a quarter and a small team, the natural next steps are: point-in-time index membership; a proper walk-forward re-estimation loop; defined-risk options overlays, which earn equity-like premia inside a 100% notional cap; execution research on the crypto leg, where 18 bp of assumed slippage is almost certainly compressible; and a search over strategy space orders of magnitude larger than one day allowed. The gap between 1.23% and 2% per month — \$39k versus roughly \$100k on the same \$10k — is worth that programme. It simply was not this programme.

Reproducibility: `make all` runs the 18 leakage tests, the backtest, the validation suite and this report from the frozen snapshot, end to end, with no network access.